Jay Doshi

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Education

B.Sc. in Computer Science & Engineering **GPA: 3.7/4.0**
University of Toledo, Toledo, OH Expected Graduation: December 2027

Relevant Coursework: Statistics, Kalman Filtering, Operating Systems, Database Systems, Software Engineering

Experience

Network Engineer Intern | Cenovus Energy, Oregon, OH **September 2026 – Present**

- Diagnosed performance degradation via root-cause analysis of SolarWinds NPM metrics, resolving issues before production outages.
- Collaborated with cross-functional engineering teams to debug connectivity defects and support production deployments.
- Integrated server, VMware virtualized, and backup systems into stable hosting, keeping plant applications highly available.

Process Automation Intern | Cleveland-Cliffs, Coatesville, PA **May 2026 – August 2026**

- Engineered a threshold-based anomaly detector on live gauge data with real-time email alerts, cutting defective plate escapes 30%.
- Automated ETL pipelines moving live sensor data from legacy OpenVMS Alpha into SQL Server, cutting manual effort by 40%.

IT Support Technician Level 2 | University of Toledo, Toledo, OH **September 2025 – May 2026**

- Resolved 40+ monthly IT tickets by troubleshooting Windows, hardware, software, and network issues, minimizing downtime.
- Deployed 150+ Windows devices via CLI tools for imaging, BIOS updates, data backups, and software installation.

Undergraduate Research Assistant | University of Toledo, Toledo, OH **May 2025 – December 2025**

- Reduced target navigation error by 40% by tuning a custom PID flight controller in a ROS2-Gazebo digital twin.
- Improved control accuracy by 60% by analyzing flight data from 15+ hardware trials in Python, computing RMSE and overshoot.
- Validated PX4 control logic in a ROS2-Gazebo simulation linking flight controller and simulator over MAVLink/UDP.
- Designed and evaluated a PPO-based reinforcement learning model for autonomous drone target navigation using ROS2-Gazebo.
- Built and configured Holybro X500 drone hardware, supporting PX4-based feedback data collection and analysis.

Professional Projects

SportsSight Analytics | Python, OpenCV, YOLOv8, Computer Vision, K-Means, NumPy **August 2025 – December 2025**

- Implemented optical flow and perspective transformation to compute real-world player speed and distance in meters.
- Fine-tuned YOLOv8 on a Roboflow football dataset, achieving 90% player and ball detection accuracy on test video clips.
- Automated team assignment by t-shirt color via K-Means pixel clustering, improving segmentation accuracy to 75%.

Multi-Agent Robotics Simulation | PX4-Autopilot, ROS 2 Humble, Gazebo Harmonic, Rviz2 **Project Link**

- Engineered a high-fidelity simulation pipeline using Gazebo Harmonic to test robotic controller logic in complex environments.
- Optimized collision physics and dynamics for realistic interaction, ensuring 95%+ parity between simulation and physical hardware.
- Built multi-agent communication protocols, enabling synchronized control and real-time data feedback.

Smart Obstacle Avoidance Vehicle | C++, Embedded Systems, Arduino UNO, TinkerCAD **Github Link**

- Developed embedded C++ software with ultrasonic sensors and servo motors for real-time obstacle detection and navigation.
- Implemented speed and distance tracking via IR sensor and rotating grid plate, achieving 85% measurement accuracy.
- Integrated Bluetooth controls, reducing command response latency by ~60% and enabling near real-time mobile operation.

Technical Skills

Programming Languages: C++, Python, C, Java, SQL, Bash Scripting, MATLAB

Robotics & Autonomy: ROS2, Gazebo, PX4, MAVLink, QGroundControl, Rviz2, PID Control, Embedded Systems, Arduino

AI & Computer Vision: Machine Learning, Reinforcement Learning, Computer Vision, OpenCV, YOLOv8, PyTorch, Deep Learning

Software Development: Data Structures & Algorithms, Algorithm Design, System Design, Unit Testing, Git, Linux CLI

Data Analysis & Tools: NumPy, Pandas, Matplotlib, Jupyter Notebook, VS Code, Docker